

# RAHUL KUMAR JHA

Machine Learning Engineer Intern | NLP · RAG · Deep Learning

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🔗 rahul5857    📁 Portfolio

## Professional Summary

AIML undergraduate with hands-on experience in Machine Learning, Deep Learning, NLP, and Retrieval-Augmented Generation (RAG) systems. Experienced in building AI-powered applications using FastAPI, LangChain, ChromaDB, Sentence Transformers, Docker, Kubernetes, and SQL. Passionate about Generative AI, semantic search, and intelligent document retrieval systems.

## Education

|                                                                                                                                                                                 |                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Bachelor of Technology (B.Tech) in Computer Science (AI-ML),</b><br><i>Malla Reddy Institute of Technology and Science</i><br>Expected Graduation: May 2027   CGPA: 8.9 / 10 | 08/2023 – 05/2027<br>Hyderabad, Telangana |
| <b>Certification Program in AI-ML with Specialization in Drone Technology, IIT Hyderabad</b>                                                                                    | 06/2025 – 01/2026<br>Hyderabad, Telangana |

## Technical Skills

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|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b><br>Python (OOP, File Handling, APIs, Data Structures, Exception Handling) | <b>Machine Learning</b><br>Regression & Classification, Feature Engineering, Model Evaluation, Cross Validation, GridSearchCV, Scikit-learn            |
| <b>Deep Learning</b><br>Neural Networks, Backpropagation, ReLU, Sigmoid, Dropout             | <b>Natural Language Processing</b><br>Sentence Transformers, Text Embeddings, Cosine Similarity, Semantic Search, Vector Search, RAG                   |
| <b>Backend Development</b><br>FastAPI, REST APIs                                             |                                                                                                                                                        |
| <b>Database</b><br>SQL (CRUD Operations, Joins, Aggregations, Query Optimization)            | <b>Data &amp; Tools</b><br>NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, Docker, Kubernetes, ChromaDB, LangChain, SQLite, VS Code, Jupyter Notebook |

## Projects

- AI Resume Analyzer | FastAPI | NLP | Sentence Transformers | Docker** [🔗](#)
- Built an AI-powered Resume Analyzer that compares resumes against job descriptions using NLP embeddings and cosine similarity.
  - Developed REST APIs using FastAPI for resume upload and automated analysis.
  - Implemented skill matching, missing skill identification, and semantic similarity scoring using Sentence Transformers.
  - Containerized the application using Docker and configured Kubernetes deployment files.
- RAG PDF Chatbot | LangChain | ChromaDB | Sentence Transformers | FastAPI** [🔗](#)
- Built a Multi-PDF RAG chatbot using Python, Streamlit, ChromaDB, and Ollama.
  - Implemented PDF parsing, text chunking, embedding generation, and vector similarity search.
  - Integrated Llama 3.2 for context-aware responses and semantic document retrieval.
  - Applied semantic retrieval using ChromaDB and LangChain to fetch relevant document chunks for question answering.
- AI-Based Landing Zone Safety Classification | AI-Assistant | Python | scikit-learn | Pandas** [🔗](#)
- Designed a classification model to assess safe drone landing zones using image-derived features.
  - Performed data preprocessing, feature engineering, and model validation to ensure accurate and reliable predictions.